

Bachelor in Computer Applications

Course Title: **Computer Graphics**

Code No: Semester: **5**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. List the major differences between DDA and Bresenham's algorithm. Illustrate the Bresenham's algorithm to the line with end points(10, 6) and (12, 1).
2. Draw the block diagram of vector and raster graphics display architecture and write the advantages and disadvantage of both display architecture.
3. How decision parameter can be used to draw circle? Calculate the points to draw a circle having radius 5 and center as (10, 5).
4. Explain how can line method detects the visible surface with example.
5. Differentiate Phong Shading with Gouraud Shading method.
6. Explain Cohen-Sutherland line clipping Algorithm.
7. What is Beizer curve? Describe its properties.
8. Discuss the use of computer graphics in different areas.
9. Describe polygon, vertex and edge table of a polygon.
10. What is boundary representation technique? Explain.
11. Explain the Phong illumination model for specular reflection.
12. Write short notes on:
a. Shearing and Scaling
b. Line Clipping